Experimental Research on MyCourses Platform

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The study investigates the challenges faced by international and American students at the Rochester Institute of Technology (RIT) while using the MyCourses learning platform. With the increasing importance of Learning Management Systems (LMS) in academic settings, the study aimed to test the hypothesis that international students encounter more difficulties using MyCourses compared to their American counterparts. The collected data, comprising responses from 21 RIT students, was analyzed to evaluate aspects such as usability, learnability, and navigation of MyCourses. The survey also collected demographic information, revealing that the majority of participants were graduate students in STEM fields, with 15 international students and 6 American students taking part. The findings hold implications for future research, suggesting the need for a more diverse participant pool, including students with prior US education experience and from non-STEM fields, to gain a comprehensive understanding of the platform's usability across different academic backgrounds. The study contributes valuable insights into the experiences of international and American students with MyCourses, highlighting the importance of further exploration of factors influencing student perceptions of LMS. This abstract provides a concise summary of the study's methodology, findings, and future research considerations, ensuring a comprehensive understanding of the study's objectives and outcomes.

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1 INTRODUCTION

Rochester Institute of Technology (RIT) has many international students from all over the world, and many are experiencing differences in education. Cultural differences are not only applied to their life but also to their educational journey. Students and faculty utilize the Learning Management System(LMS) called MyCourses at RIT, which plays an essential role in the classroom. We recruited more than 20 participants and utilized an online survey to learn about their experience with myCourses. With the growing popularity of LMS, increasing numbers of education institutes opted for this alternative system to help students through their academic journey. [3] Understanding such importance along with the negative results from previous studies [4], we designed a survey to investigate students' interaction with myCourses based on the hypothesis International students would experience more difficulty using Rochester Institute of Technology's learning platform, MyCourses than students from the United States. In this paper, we look into the data we collected through the online survey and analyze it to find out whether our hypothesis is true or not.

2 METHODS

2.1 Data collection

We used a Google Form online survey consisting of three demographic questions: 15 about the international student experience and 13 about the US student experience. Our hypothesis compares the experiences of international students and US students. Here, the categorical variable demonstrates the student's demographic

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status, namely whether the student is an "International" student or a "United States" student. This variable is being manipulated or used as a grouping factor to compare the different experiences of each group with the learning platform. It is the independent variable as it is not influenced by any other variables in the study. Similarly, the difficulty experienced in using the Rochester Institute of Technology's learning platform, which can be measured by various aspects such as usability, learnability, and navigation score of the learning platform. These are outcome variables that are expected to possibly change based on the students' demographic status (international vs. US). These variables are dependent on the independent variable and may be influenced by it. We created two different sections in the form that appropriately ask our participants questions. Here is the list of questions we asked our participants, starting from the demographic questions: What year are you? What is your major? Are you an international student? Participants were given different sets of questions based on the answer to question 3. If participants answered yes to question 3, they took section two, which is a set of questions for international students. If participants answered no to question 3, we provided the questions seeking the usability and experience of American students. The list of questions is in the appendices below.

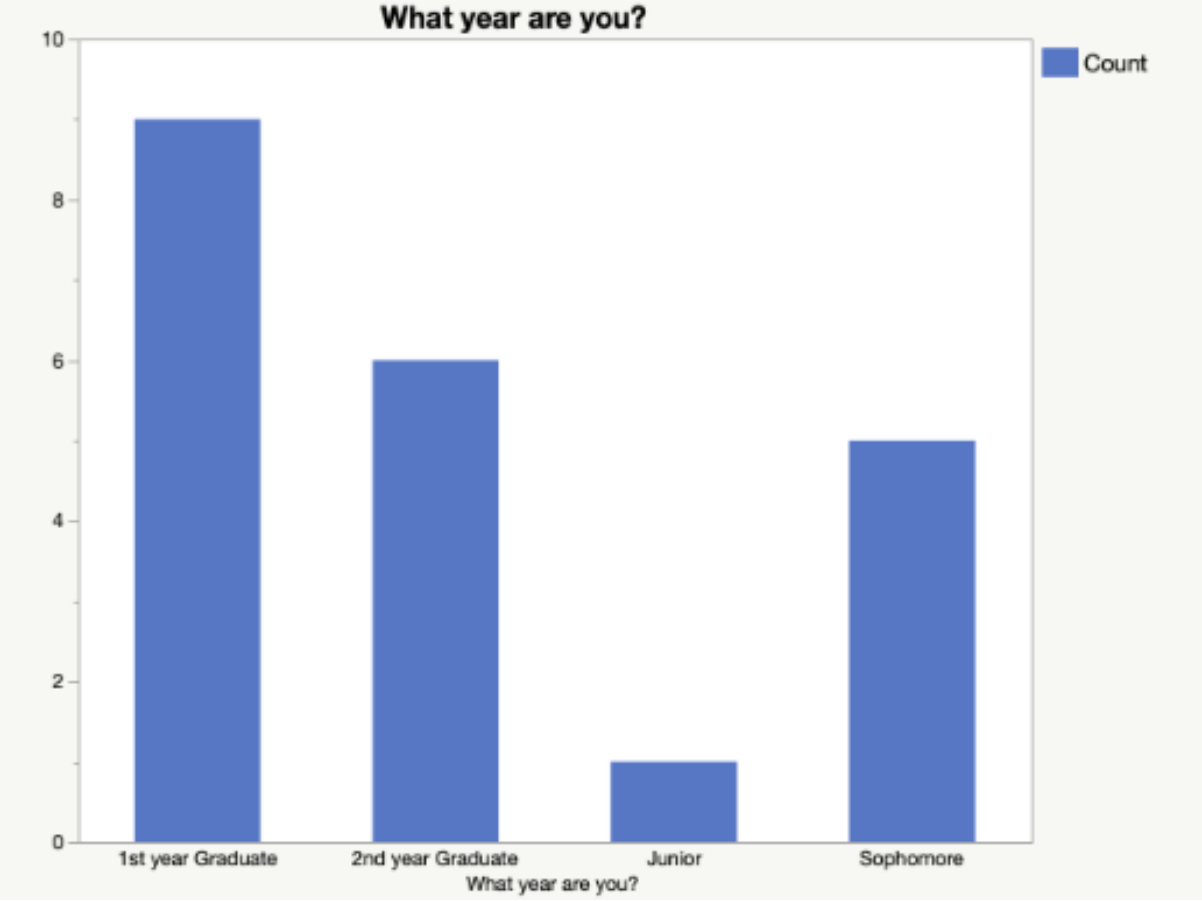


Fig. 1. Participants Year of study

2.2 Participants Demographic

We have successfully recruited 21 RIT students who are currently seeking degrees, and as you can see in Figure 1 confidentially, the majority of our participants are graduate students, some sophomores and one junior. Their study fields are mostly in the STEM field, including Human-Computer Interaction and Data Science. We have 15 international students and six US students who participated in our study. None of the international students have prior US educational experience. All of our international student participants are from India. Some of them speak English as their first language, but many of them speak several languages, including Hindi, Telugu, and Marathi.

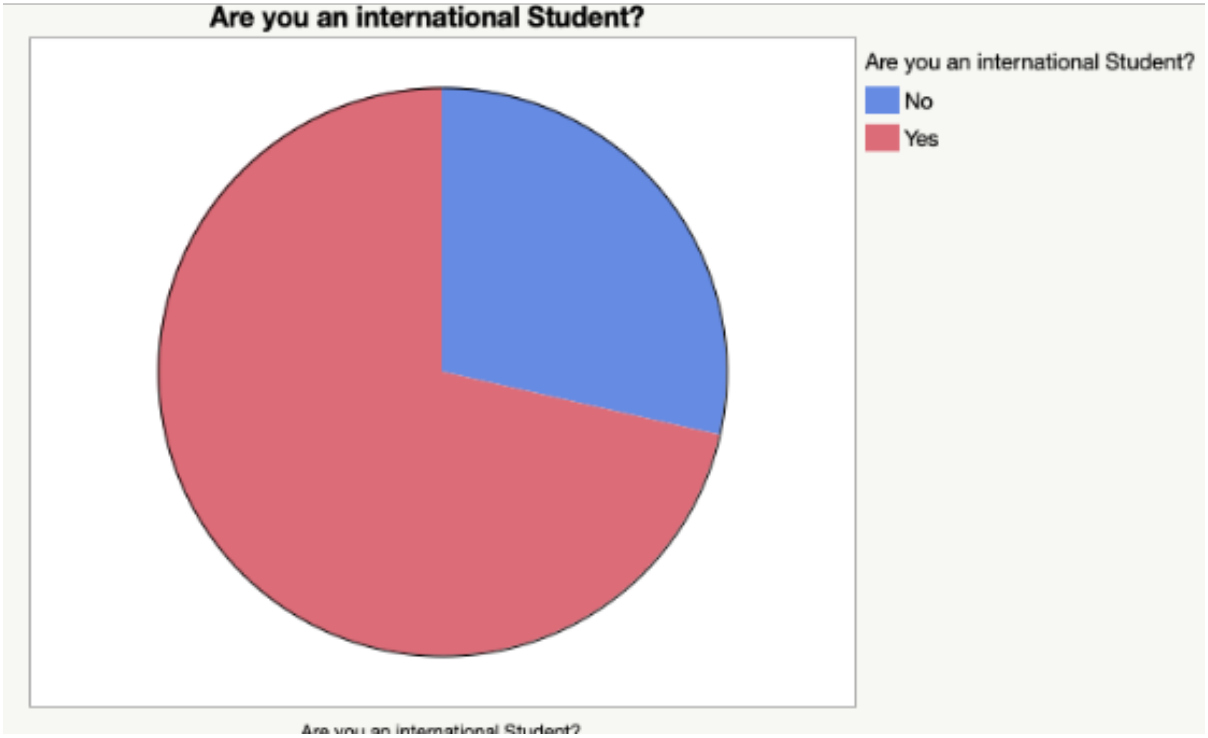


Fig. 2. International vs US students

2.3 Note on Participant Engagement and Response Data

We did not have the opportunity to track the total number of people who clicked on the link to our Google Form. As a result, we cannot provide specific data regarding the response rates, which will detail how many individuals we reached out to versus how many participated in the survey. We based our analysis and conclusion on the responses we received without the context of the overall engagement levels.

3 RESULTS

We collected the data with a separate set of questions based on the demographic information. We used JMP Pro to visualize and analyze the data we have collected from our participants. We had four major findings listed below; Experience and Usability, Learnability of MyCourses, Similarity to Other Learning Platforms and Assignments and Content Navigation. We listed each data separately between international students and American students and used a scale of 1 to 5 to rate the aspects based on 1 being best and 5 being worst. We have rounded up with two digits and provided you the calculation of mean, standard deviation and any other statistical results below.

3.1 Experience and Usability

40% of our international participants and 66.7% of our American participants used other learning platforms, such as Google Classroom, and most international students have been using MyCourses for about a year or less. On the other hand, a couple of American students have been using MyCourses for over two years. International students rated a mean of 3 about the usability of MyCourses with a standard deviation of 0.93. American students rated the usability of MyCourses as a mean of 3.2 and a standard deviation of 0.75. Some international students shared their both positive and negative experiences that "Yes, was not able to find out deadlines until I do small deep dive and enter that particular assignment." "It wasn't confusing. Just there are a few features which I didn't know before for example, option to send email reminders for assignment deadlines and for announcements or change in content." "I wish some of the functions are easy to discover" American students also shared both positive and negative experiences that "There's a lot of sections, I still get confused and sometimes miss assignments." "Only confusing thing was going back to things using the buttons on the page."

3.2 Learnability of MyCourses

We found out the usability and first-time experience; then we collected the learnability of MyCourses which is the rate and impression of how users could learn based on their first time. International students rated a mean of 2.26 for learnability of MyCourses and a standard deviation of 0.96. American students rated a mean of 2.33 for learnability of MyCourses and a standard deviation of 1.21. Along with the rating, international students shared the story and the reason for rating: There was nothing much to learn other than how to navigate around to check what the due dates were, how to check the assignment description and locate the posted documents. takes some time to get used to the UI as well to remember the correct spots for the action American students also shared similar experiences: Its not difficult to learn how to navigate, just tedious

3.3 Similarity to Other Platforms

As we mentioned above, roughly half of the participants used other learning platforms, such as Google Classroom, Edex, and Coursera. We collected the rate of how similar MyCourses is to these platforms. Six international students have used other learning platforms before, and the rate for the similarity to these platforms is a mean of 3.81 and a standard deviation of 1.16. Four American students have used other platforms, and the rate of similarity is a mean of 3.66 and a standard deviation of 1.21. International students shared their other learning platform experiences and compared to MyCoursesL Google Classroom has a completely different interface in terms of Card Sorting, also, the professor had all the capabilities to modify the classroom activities integrating all the google suite applications that works on the browser itself. American students also shared some thoughts about the comparison between MyCourses and other products: They are pretty similar. They have course shells and you submit assignments on there, no quizzes though

3.4 Assignment and Content Navigation

We looked into the student's impression about the content organization of MyCourses. International students have responded to the rate of content organization with a mean of 2.33 and a standard deviation of 0.89. American students rated the content organization with a mean of 2.88 and a standard deviation of 0.75. We have similar feedback from both international and American students, such as there are so many sections scattered that they missed the assignments.

4 DISCUSSION

Based on the findings of MyCourses, we analyzed the collected data using inferential statistical analyses to evaluate the user-perceived aspects of usability, learnability, and navigation. We used a t-test for two sample means to compare and tried to prove our hypothesis in each finding with an assumption of unequal variance. Consistently, the null hypothesis is that international students and American students have an equal or similar mean and the alternative hypothesis is there is a significant difference between international students and American students.

4.1 Usability of MyCourses

We have collected the usability rating results from two groups, international and American students, and we have generated mean and standard deviation for both groups in the results. Based on these results, we utilized a t-test and generated a score of 0.51, with a critical value of +/-2.19, as we have attached Figure 4. We sought the difference between the two groups, which was that international students may struggle more and have a worse impression of MyCourses in terms of usability. In other words, we looked into the cultural and educational background differences as a factor in this section. A p-value of 0.6170 was within the critical region; we failed to reject the null hypothesis, which resulted in the fact that students' demographics based on their home countries are not a factor in the usability of MyCourses.

4.2 Learnability of MyCourses

A t-test was used for learnability as well and generated a score of 0.12, with a critical value of +/-2.20, as we have attached Figure 5. We sought the difference between the two groups, which was that international students may struggle more and have a difficult time learning how to use MyCourses. A p-value of 0.90 was within the critical region, which leads to the observation that learnability as a factor is not affected as well.

4.3 Navigation of MyCourses

We have collected data about the content organization and navigation. This section provided us with a deeper understanding of usability and learnability. Students always look for assignments, quizzes, and lectures on LMS; in other words, content findings are major tasks for students and affect a lot of usability ratings. These content findings were difficult based on the previous study [4]. The score for navigation was 1.43, with a critical value of +/-2.20, as in Figure 6. Therefore, we failed to reject the null hypothesis and conclude that there is no significant difference in content-finding issues between the two groups. We could not find any differences in usability and content findings between international and American students; however, we have had multiple feedbacks about the complications, and it is hard to find some they were looking for. This resulted in missing some deadlines and assignments. Based on that, technical support with necessary redesign may help solve student struggles [1].

4.4 Similarity

We looked at the comparison to other learning platforms, such as Google Classroom. These may help us look into the differences and comparisons between two groups, specifically a human factor. These two different groups

of educational background differences may impact how they perceive the usability and learnability of LMS. As you see in Figure 7, there are no differences between the two groups, which could possibly provide the fact that human perception differences are very minimal, and the above three statistical results comparing international and American students with MyCourse could be trusted.

4.5 Limitations

This study faces several limitations that may influence the interpretation and generalizability of the findings. First, the sample size of the study was relatively small, comprising just over 20 participants. While this number allowed for an initial exploration into the usability and learnability of MyCourses at Rochester Institute of Technology, the limited sample size restricts the statistical power of the analyses conducted. Moreover, the distribution of participants between the two groups—international and US students—was unequal. This disparity in group sizes may introduce biases that affect the reliability of comparative analyses. Unequal group sizes can lead to skewed results that might not accurately reflect the true nature of any existing differences between international and domestic students' experiences with MyCourses. Furthermore, the study's reliance on self-report measures can be subject to response biases, such as social desirability or recall bias, which might impact the accuracy of the data collected. Despite these limitations, the study provides valuable insights into the experiences of a diverse student body with MyCourses platform.

5 CONCLUSION

In conclusion, we designed a survey to investigate students' interaction with myCourses based on the hypothesis that international students would experience more difficulty using Rochester Institute of Technology's learning platform, MyCourses, than students from the United States. We successfully recruited 21 RIT students across the campus and collected qualitative and quantitative data, such as usability, learnability,, and navigation of MyCourses, listed in the result section. Based on the result, we deeply looked into and provided the results. The survey findings revealed significant insights into the experiences of international and American students with MyCourses. Contrary to the initial hypothesis, the study did not find evidence to support the idea that international students faced more challenges with the platform. The reason may be the globalism and LMS is getting famous and common tool in the education field. [2] Looking ahead, future research could involve a more extensive and diverse participant pool to explore how factors such as prior US educational experience and field of study might influence student perceptions of MyCourses. Additionally, including participants from non-STEM degrees could provide a more comprehensive understanding of the usability of MyCourses across different academic fields.

6 FUTURE WORK

In the future, we hope to have more diverse participants. For example, we wonder if international students with prior US education experience may have different impression and perception to the usability of LMS. On top of that, students from non-STEM degrees may impact the result of our study as we have recruited most of them from STEM degrees in this study.

REFERENCES

[1] Nor Adzharuddin. 2013. Learning Management System (LMS) among University Students: Does It Work? *International Journal of e-Education, e-Business, e-Management and e-Learning* (01 2013). https://doi.org/10.7763/IJEEEE.2013.V3.233

[2] Peter Kpolovie and N Lale. 2017. GLOBALISATION AND ADAPTATION OF UNIVERSITY CURRICULUM WITH LMSs IN THE CHANGING WORLD. 5 (05 2017), 28–89.

[3] Kung Siew Min, Fadhilah Mat Yamin, Wan Hussain, and Wan Hussain Wan Ishak. 2012. The Usage of LMS among Undergraduate Students. https://api.semanticscholar.org/CorpusID:740228

[4] Yugo Iwamoto Steve Chen and Damionie Vernon. 2024. Usability Research of Rochester Institute of Technology's Learning Management System. (2024).

7 APPENDIX

7.1 Survey questionnaire

Here is the list of questions we asked our participants, starting from the demographic questions:

(1) What year are you?
Freshman, sophomore, junior, senior, 5th year, 1st grad, 2nd grad and other

(2) What is your major?

(3) Are you an international student?
Yes or no

Participants were given different set of questions based on the answer of question 3.

—— Page change —– Section 2
If participants answered yes to question 3, they have been asked these questions below which is for international students.

(1) What is your nationality?

(2) What language do you speak other than English? Is English your first language?

(3) Do you have prior US education experience?
Yes or no

(4) If you have prior US education experience, please list the school and degree you worked on prior to RIT.

(5) Have you ever used a platform like myCourses before?
Yes or no

(6) How similar was it to myCourses?
Likable scale from 1 to 5

(7) Please rate the usability of MyCourse.
Likable scale from 1 to 5

(8) Can you explain your first experience using MyCourse? Was it confusing? If so, what made you confused?

(9) How long did it take to get used to MyCourse?
Immediate, a few days, a week, more than a week, a month, more than a month, still confusing to me.

(10) How do you rate the learnability of MyCourses?
Likable scale from 1 to 5

(11) Please explain why you rated the learnability of MyCourse here

(12) How would you rate the assignment and content navigation provided by myCourses?
Likable scale from 1 to 5

(13) Have you ever used any other learning platforms? If so, please list the product.

(14) Any other feedback and comments about the usability and learnability of MyCourses?

(15) If you have used other learning platforms, please compare the products to MyCourse and explain how they are different here.

—— Page change —– Section 3
If participants answer no on demographic question 3, this is for US students.

(1) Have you ever used a platform like myCourses?
Yes or no

(2) If you have used others, how similar was it to myCourses?
Likable scale from 1 to 5

(3) How long have you been using the MyCourses platform?
Less than six months, six months to a year, 1-2 years, more than two years

(4) Do you consider MyCourse as easy to use?
Yes or no

(5) Please rate the usability of MyCourse.
Likable scale from 1 to 5

(6) Can you explain your first experience using MyCourse? Was it confusing? If so, what made you confused?

(7) How long did it take to get used to MyCourse?
Immediate, a few days, a week, more than a week, a month, more than a month, still confusing to me.

(8) How do you rate the learnability of MyCourses?
Likable scale from 1 to 5

(9) Please explain why you rated the learnability of MyCourse here

(10) How would you rate the assignment and content navigation provided by myCourses?
Likable scale from 1 to 5

(11) Have you ever used any other learning platforms? If so, please list the product.

(12) Any other feedback and comments about the usability and learnability of MyCourses?

(13) If you have used other learning platforms, please compare the products to MyCourse and explain how they are different here.

7.2 Analysis References

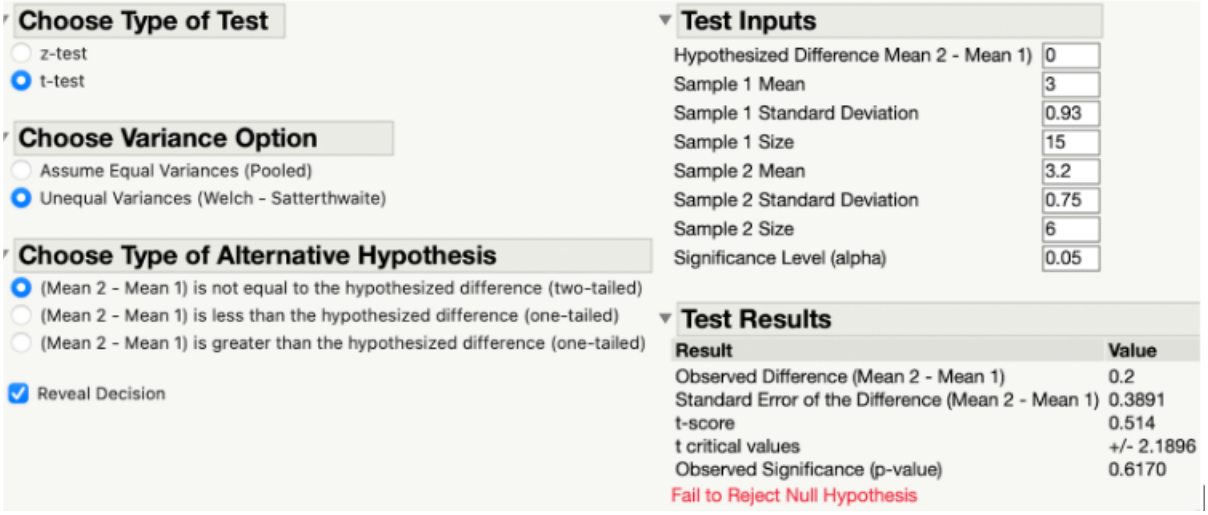


Fig. 3. t-test on Usability

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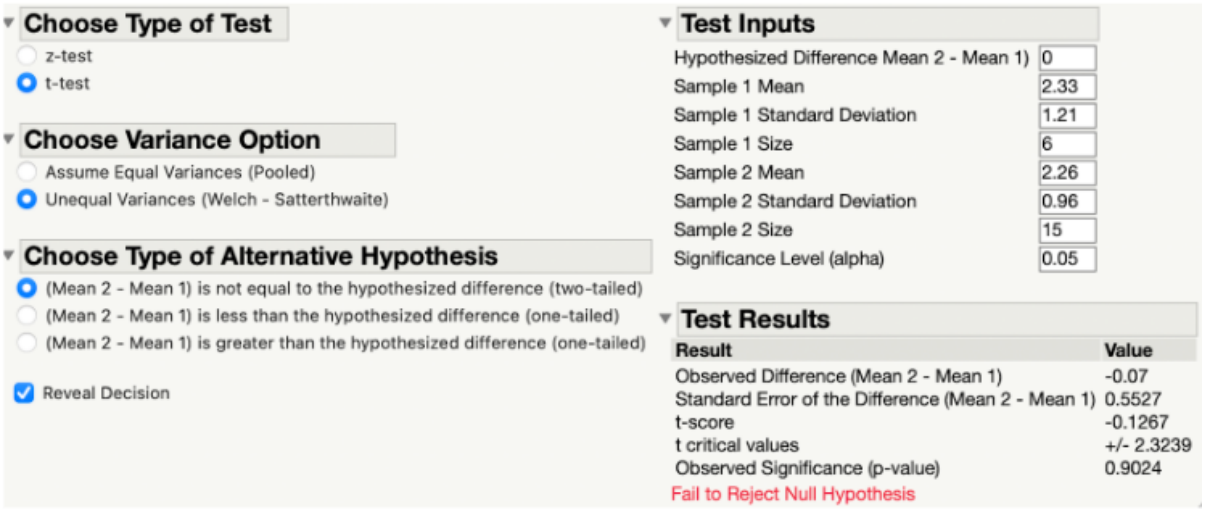


Fig. 4. t-test on Learnability

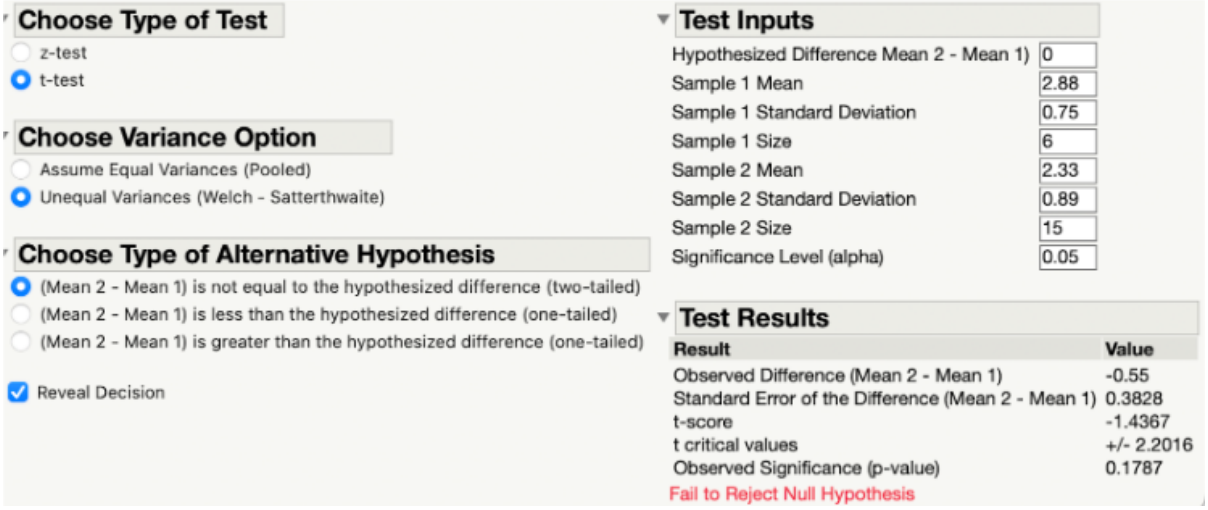


Fig. 5. t-test on Navigation

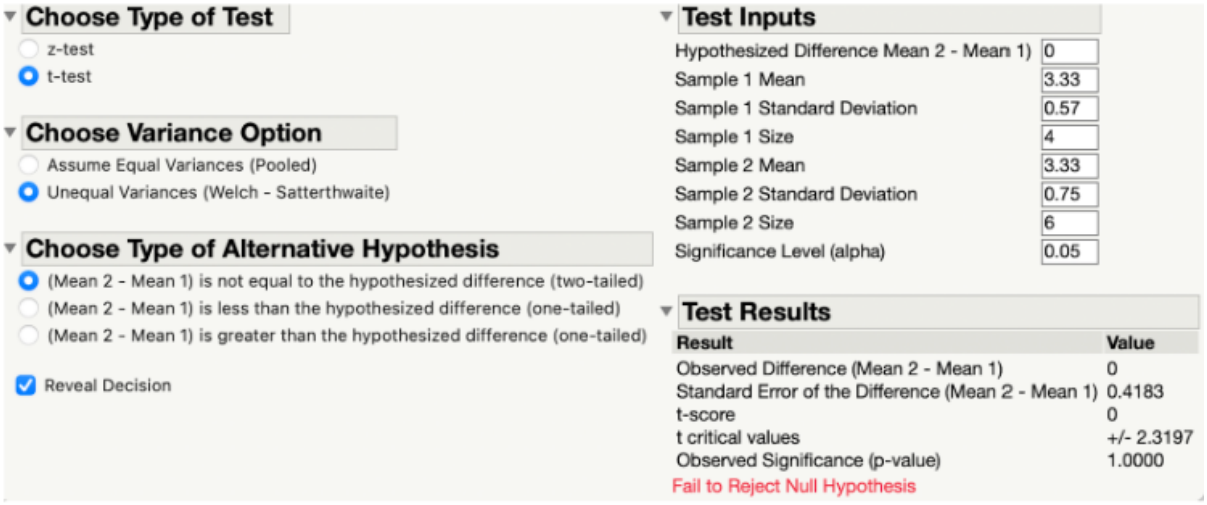


Fig. 6. t-test on Similarity